IBM Innovation Centre for Education

ICE Day 2026 Hackathon

Track 1: Sustainable AI & Green Computing (3 Statements)

1. GreenPrompt: LLM Token & Carbon Footprint Optimizer

- **Core Problem:** Running heavy frontier LLMs for trivial queries consumes unnecessary compute and power.
- **Workable Scope:** A Python/Streamlit tool that intercepts user prompts, evaluates prompt complexity using a lightweight heuristic (or small BERT/sentence-transformer model), routes simple queries to a compact local/SLM model (e.g., Llama 3 8B or lightweight API), and quantifies estimated token cost and watt-hours saved.
- **Judging Deliverable:** Live prompt playground demonstrating dynamic routing and an energy savings dashboard.

2. EcoLens: Smart Circular Waste Classifier

- **Core Problem:** Mixed municipal and office waste frequently ends up in landfills because segregation guidelines are inconsistent and confusing.
- **Workable Scope:** A web app (MERN or Streamlit) where users upload or capture a photo of trash (plastic bottle, e-waste, compostable container). Using a vision API (e.g., Gemini Vision API or a pre-trained MobileNet), the app classifies the item, gives localized recycling disposal steps, and logs a personal "diversion from landfill" score.
- **Judging Deliverable:** A working camera/file upload interface that correctly classifies 4–5 sample items in real time.

3. WattWatch: Building Energy Anomaly & Peak Alert Dashboard

- **Core Problem:** Commercial and lab spaces run HVAC, lighting, and heavy hardware continuously without detecting idle energy leaks.
- **Workable Scope:** A Python backend parsing public building time-series energy consumption data (Kaggle dataset), running an anomaly detection algorithm (Isolation Forest or simple rolling-z-score), and flagging overnight energy spikes with an estimated carbon emission penalty.
- **Judging Deliverable:** A lightweight interactive visual dashboard showing power consumption curves with flagged anomalies and simulated alerts.

Track 2: Cybersecurity & Defensive Tech (3 Statements)

4. PhishGuard: GenAI-Enhanced Spear-Phishing & Quishing Detector

- **Core Problem:** Attackers now generate highly customized spear-phishing emails and malicious QR codes ("quishing") that bypass basic spam filters.
- **Workable Scope:** A Python service that takes raw email text/headers or uploaded QR images, extracts URLs, decodes obfuscated links, runs heuristic domain-age/spoof checks

(e.g., using VirusTotal or open threat feeds), and uses an LLM to explain *why* the message is suspicious.
- **Judging Deliverable:** A UI where judges paste a phishing sample and receive a risk score (0–100) alongside highlighted red flags.

5. LogSentinel: Lightweight Real-Time SSH & Web Log Analyzer

- **Core Problem:** Server administrators get inundated with thousands of raw log lines daily and miss credential stuffing or brute-force patterns.
- **Workable Scope:** A Python CLI or mini-dashboard that ingests a simulated Apache/Nginx or `auth.log`stream, identifies repeat IP bursts, parses failed attempts, correlates IP geographic origins, and outputs an automated ban list or firewall rule rule-set (`iptables`/UFW syntax).
- **Judging Deliverable:** Ingesting a sample 10,000-line server log and instantly showing top attacker IPs, attack vectors, and exported firewall rules.

6. SafePush: Automated Secret Leak & API Key Sanitizer

- **Core Problem:** Junior developers frequently commit sensitive `.env` secrets, AWS keys, and database credentials directly to public repositories.
- **Workable Scope:** A pre-commit simulation or GitHub webhook engine built with Node.js/Python that uses regex and Shannon entropy detection to flag exposed secrets before code is committed, generating a masked, clean replacement and a git-history remediation script.
- **Judging Deliverable:** Live commit/paste test showing immediate redaction and an audit report of discovered API keys.

Track 3: AI / Machine Learning & Applied NLP (3 Statements)

7. DocuRAG: Offline/Lightweight Contextual Research Assistant

- **Core Problem:** Employees spend hours searching through dense technical manuals, bylaws, or academic syllabi for exact clauses.
- **Workable Scope:** A Retrieval-Augmented Generation (RAG) prototype built using LangChain/LlamaIndex, FAISS/ChromaDB, and a free-tier LLM API. The user drops a 10–20 page PDF, and the system answers questions with highlighted exact-page citations.
- **Judging Deliverable:** Uploading a sample policy document, asking 2 specific edge-case questions, and seeing answers with source snippet attribution.

8. FairHire: Resume Bias & Anonymization Engine

- **Core Problem:** Unconscious bias based on name, gender markers, age, or postal address skews early-stage recruitment screening.
- **Workable Scope:** A Python NLP tool using spaCy or an LLM that ingests PDF resumes, automatically redacts Personally Identifiable Information (PII) and demographic signals,

and ranks the anonymized candidate against a job description strictly on relevant technical keywords and project impact.
- **Judging Deliverable:** Side-by-side view of raw vs. sanitized resume with candidate match scores.

9. SynthVoice: Real-Time Audio Deepfake & Synthesized Speech Flagger

- **Core Problem:** Voice cloning scams are rising rapidly across social messaging apps and phone calls.
- **Workable Scope:** A Python Streamlit app using a pre-trained audio classification model (or Librosa-extracted spectral flatline/zero-crossing rate features via a simple classifier) to evaluate uploaded 5-second WAV voice clips and output an authenticity confidence probability.
- **Judging Deliverable:** Uploading two audio clips (one recorded human, one cloned/TTS) and viewing audio spectral differences and confidence scores.

Track 4: Core Python & Automation (3 Statements)

10. DataDoctor: Automated CSV Profiler & Data Cleaning Pipeline

- **Core Problem:** Data analysts spend 60% of their time fixing formatting inconsistencies, outliers, and missing values before analysis.
- **Workable Scope:** A Python application that takes any messy CSV/Excel dataset, auto-detects date formats, fixes corrupted headers, flags outliers via statistical box-plots, imputes missing values, and exports a clean dataset plus an automated 1-page PDF summary report.
- **Judging Deliverable:** Uploading a deliberately corrupted dataset and generating a clean CSV ready for training in under 10 seconds.

11. PulseScrape: Multi-Source Sentiment & Trend Aggregator

- **Core Problem:** Product teams struggle to quickly track live customer sentiment across dispersed public forums when launching new features.
- **Workable Scope:** A Python scraping/API pipeline (using Reddit API, RSS feeds, or mock e-commerce reviews) that pulls the latest 100 comments on a keyword, runs VADER/TextBlob sentiment analysis, and visualizes positive/negative shifts with word clouds and trend line charts.
- **Judging Deliverable:** Typing a search term (e.g., "iPhone battery life") and rendering an auto-generated live breakdown of public opinion.

12. AutoDoc: Python Repository to API Spec & Architecture Visualizer

- **Core Problem:** Undocumented legacy codebases slow down team onboarding and feature expansion.
- **Workable Scope:** A Python CLI tool using `ast` (Abstract Syntax Trees) to scan an uploaded directory of Python scripts, extract all functions, classes, decorators, and docstrings, and automatically render an interactive architecture diagram and a Markdown API reference.
- **Judging Deliverable:** Running the tool against an open-source mini-project and displaying the generated dependency graph and API docs.

Track 5: Full-Stack MERN Development (4 Statements)

13. PeerSkill: Time-Bank Skill Barter Exchange

- **Core Problem:** Students and freelancers want to learn skills from peers (e.g., "teach me React, I'll teach you Figma") without exchange of money.
- **Workable Scope:** A MERN stack platform where users post skills they offer and skills they want. A credit system tracks 1 hour given = 1 credit earned. Features include basic user profiles, a live search/filter catalogue, and a booking request modal.
- **Judging Deliverable:** Seamless end-to-end user flow: User A lists a skill, User B requests a session, and credits transfer upon acceptance.

14. EventPulse: Micro-Community Event Ticketing with Dynamic QR Check-in

- **Core Problem:** College festivals and local meetups struggle with clunky ticketing setups and long on-ground entry queues.
- **Workable Scope:** A MERN application where organizers create events, attendees generate unique registered tickets with signed JWT-based QR codes, and a mobile/web camera scanner mode allows organizers to validate tickets and prevent double entry.
- **Judging Deliverable:** Generating a ticket on a laptop screen and scanning it live with a phone camera to mark the attendee as "Checked In."

15. AgriDirect: Micro-Farmer Direct-to-Consumer Market Board

- **Core Problem:** Small-scale regional producers lose margins to layers of local middlemen because they lack simple listing boards.
- **Workable Scope:** A localized MERN marketplace with a simplified inventory input (product, harvest date, location, price per kg) and a WhatsApp direct-chat integration (`wa.me` links) for buyers to confirm orders directly without payment gateway bottlenecks.
- **Judging Deliverable:** A buyer browsing fresh produce listings filtered by proximity and clicking to open a pre-filled WhatsApp order chat.

16. OpsBoard: Team Incident Management & Escalation Wall

- **Core Problem:** Small engineering teams need an intuitive, real-time command dashboard when production systems go down.
- **Workable Scope:** A MERN stack real-time status board using Socket.io. Team members can file an incident (P1 to P4), assign responders, update incident status ("Investigating" -> "Identified" -> "Resolved"), and log a timestamped timeline of mitigation actions.
- **Judging Deliverable:** Opening the dashboard in two separate browser windows and observing real-time state synchronization when an incident status updates.